



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Information System's Vulnerability



Learning Outcome:

Identify key information about people's jobs and places of work in spoken or written contexts.

Design the technological architecture of the IT system, through hardware and software recognition, based on available technology in the market, the analysis report and the distribution diagram.

Can Do Statements:

- I can explain the concept of vulnerability.
- I can give details to avoid vulnerability.
- I can explain what things need to be modified in a business to avoid vulnerability.

Ask your partner:

- What computer viruses or system information threats have you had to deal with at home or at work?
- What measures have you taken to avoid your PC from computer bugs and any other threats?





Let's Explore!

The IT officer of a company is talking to the general manager about dealing with system vulnerability and data loss.



Instructions: Read the text below.

General Manager:

We're having lots of problems with our information systems and computers nowadays. Data losses, computer viruses, spyware, and a system attack have been very common. I won't wait here inactively to see how our information assurance gets damaged or is put at risk. So, what alternatives do we have?

IT officer:

I'm used to this sort of computer threats all the time, Mrs. West. So, based on the collected information and analysis so far, my team's working on implementing a series of measures to avoid the system susceptibility to fail and prevent it from any form of threat. For example, in my previous job I was used to the assessment of an Information Technology Security Audit that tested us on how well we were working on security controls. Hopefully, we'll implement that functionality by the end of this month. In addition, we're working on the implementation of an Information Security Management System which will operate at its full capacity in about three weeks and ...

General Manager:

Sorry, but I don't understand anything.

IT officer:

Oh, excuse me, Mrs. West. What I'm saying is that I'm used to working with and consequently, we're implementing plans and standards to avoid any vulnerabilities and risk for any of the company's information systems and assets. As you probably know, a vulnerability is a flaw or weakness in the design, implementation, operation or management of a system that could be exploited by a hacker's capability to violate the system's security procedures and controls.

General Manager:

And when will we see results?

IT officer:

A sensibility program will start next Thursday to let you know what things will be modified in the business to avoid vulnerability. That relates to management procedures, hardware, software, personnel, physical environment, communications equipment, and business operations.



**Let's Practice!**

Follow my advice!

Instructions: Read the following sentences based on the text. In front of each sentence write True or False.

Well, to avoid vulnerabilities you will have to implement plans and standards. Those plans will help your company to avoid risk for any of the company's information systems and assets. In other words, you will improve the design, implementation, operation and management of the system. That way, no one will violate the system security procedures and controls.

1. Plans and standards will help the vulnerabilities. _____
2. They will improve all aspects of the information systems in the company. _____
3. Security controls and procedures won't be violated. _____

**Let's Practice!**

Let's work on the problem!

Instructions: Find the missing words for each sentence.

CLUES

1. We are used to having problems with the information s _ _ _ _ and computers.
2. I'm used to seeing everything in order, I don't want our information assurance is at _ _ _ .
3. I'm used to treating this kind of _ _ _ _ _ and solve them in the best way.
4. I hope you can do a really god job in a short time because we are used to solving these _ _ _ _ _ problems really fast.
5. I'm used to working under _ _ _ _ _ .
6. The company is used to implementing plans and _ _ _ _ _ to avoid any vulnerabilities.

P	Q	W	A	L	G	L	E	V	X	Q	S	H	F	Z	N	X
R	S	D	M	G	B	I	B	E	D	B	J	F	I	H	R	F
K	F	S	Y	S	T	E	M	Z	X	O	P	N	Y	V	T	Q
W	M	X	M	I	Q	R	H	T	M	P	C	D	J	U	P	C
A	X	W	H	G	Y	X	A	Q	L	R	M	R	T	L	P	E
S	S	W	G	F	K	F	B	K	J	E	E	Z	X	N	D	I
L	T	P	R	F	Y	H	B	D	A	S	S	Z	Y	E	G	Z
H	L	P	I	L	V	L	G	J	F	S	T	U	J	R	A	S
D	V	C	S	Y	Q	B	X	K	H	U	Q	O	Y	A	U	X
B	I	G	K	T	D	G	A	Y	O	R	J	U	X	B	W	X
P	S	L	V	X	N	N	J	J	D	E	I	A	I	I	Q	J
A	E	E	L	A	I	M	O	K	N	Z	A	V	E	L	C	C
Z	H	M	P	R	O	B	L	E	M	S	D	L	W	I	L	X
N	G	O	F	O	C	T	T	D	H	C	H	U	M	T	L	S
A	X	S	T	A	N	D	A	R	D	S	V	S	Y	Y	Z	I
J	P	E	F	Y	W	W	Y	J	Z	K	Y	F	S	W	A	M
Z	Y	F	T	H	S	X	P	I	C	L	W	A	X	W	O	L



Extension Activity

Looking for Advice About Vulnerability Issues!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about vulnerability.
- Identify the best practices companies should implement to avoid vulnerability.
- Make a list of things companies are used to doing to avoid vulnerability.
- Share your list with your class and discuss.

Self-Assessment

Yes

No

- I search for information about vulnerability.
- I identify the right practices companies implement to avoid vulnerability in their businesses.
- I classify the information I got into common behaviors companies are used to doing to avoid vulnerability.
- I share information and discuss about the topic.

☐ ☐
☐ ☐
☐ ☐
☐ ☐



Target Words

- **computer virus** (noun): a small computer program or a programming code that can make copies of itself and is able to attack a computer or network system through data files to prevent it from working well
e.g. We've had lots of problems with our information systems and computers recently. Data losses, viruses, spyware, and a system attack have been very common.
- **information assurance** (noun): is a series of procedures and steps implemented to maintain integrity, availability, authenticity, and confidentiality of data in an information system
e.g. I won't wait here inactively to see how our information assurance gets damaged or is put under risk.
- **computer threat** (noun): any possible form of danger that might exploit a system information's vulnerability and so cause harm
e.g. I'm used to this sort of computer threats all the time, Mrs. West.
- **susceptibility** (noun): the condition of being easily infected, harmed, or influence by something for having a weakness or flaw in its system
e.g. My team's working on implementing a series of measures to avoid the system susceptibility to fail and prevent it from any form of threat.
- **business asset** (noun): something valuable on which a company has ownership and serves to produce cash and profits
e.g. We're implementing plans and standards to avoid any vulnerabilities and risk for any of the company's information systems and assets.
- **personnel** (noun): all the employees, the people who work in an organization
e.g. That relates to management procedures, hardware, software, personnel, physical environment, communications equipment, and business operations.



Learning Points

Be Used to + Ing Form to Talk About Customs

Be used to is an expression, not a tense. However, it can be employed in the present and past forms. We use it to say we are familiar with or accustomed to something.

Grammatical Structure

Be used to is very simple to employ. We simply need to change the tense of the verb "to be" depending on the subject it goes along. In case we need a verb after the expression "used to", that verb must have the -ing form. Look at the structures:





Learning Points

1. Subject + **is/are/was/were** + **used to** + **verb -ing** + complement
2. Subject + **is/are/was/were/will be** + **used to** + **noun** + complement

Examples:

Present

- ***I'm used to checking** my PC every day for viruses or any other threat as the first thing I do when I arrive to my office.*

Past

- *I preferred my old office. I **was used to its big windows** that provided lots of natural light. My actual office is much dark*

Examples

- *I'm used to this sort of computer threats all the time, Mrs. West.*
- *In my previous job I was used to the assessment of an Information Technology Security Audit that tested us on how well we were working on security controls.*

Data Loss

Data loss is the damaged or destruction of information due to failures in the storage, transmission and processing processes within an information system.

To avoid data loss from happening companies implement robust equipment and security processes and protocols to back up and restore information. Two concepts related to this topic are usually explored. They are:

- data loss and (permanent loss of information)
- data unavailability (temporary loss of information)

Companies adopt or design and develop security systems to avoid data losses primarily. Data unavailability may be caused due to a network outage which is a temporary cause. However, it should be included in the risk analysis plan that may affect the system.

Types of Data Loss

- Caused by intentional action: Intentional file or program deletion
- Caused by unintentional action: accidental file or program deletion, administration errors, inability to read unknown file format, misplacement of CDs or Memory sticks



Learning Points

- Caused by failure: power failure, hardware failure, a software crash or freeze. Software viruses or reduced usability, such as not checking a file delete command. Business failure, and data corruption, such as file system corruption or database corruption.
- Caused by a disaster: earthquake, flood, fire, natural disaster, tornado, etc.
- Caused by crime: SQL injection, hacking, theft, sabotage, a malicious act, such as a worm, virus, hacker or theft of physical media.

Cost of Data Loss

Data that is needed but unavailable because of data loss is highly expensive. The cost of a data loss incident is directly related to the value of the data and the span of time that it is unavailable. You may think about:

- The cost of keeping working without the data
- The cost of creating the data once again
- The cost of informing users in the occurrence of a compromise

Prevention of Data Loss and Data Unavailability

Companies can take a series of preventive measures to avoid data loss and data unavailability from occurring:

- Installation of multiple power circuits with battery backup and a generator to be protected against power failures.
- Installation of an uninterruptable power supply to protect computer's drives against sudden power spikes.
- Installation of a journaling file system and RAID storage to be protected against software and hardware failure.
- Ensuring minimal vibration and movement will protect hard disk drives against damaging the internal components.
- Ensuring regular data backups to help recover more easily after a data loss event.
- Ensuring security plans and protocols to prevent the system from user errors, natural disasters and criminal events.

Examples

- *We've had lots of problems with our information systems and computers recently. Data losses, computer viruses, spyware and a system attack have been very common.*
- *In addition, we're working on the implementation of an Information Security Management System which will operate at its full capacity in about three weeks.*

